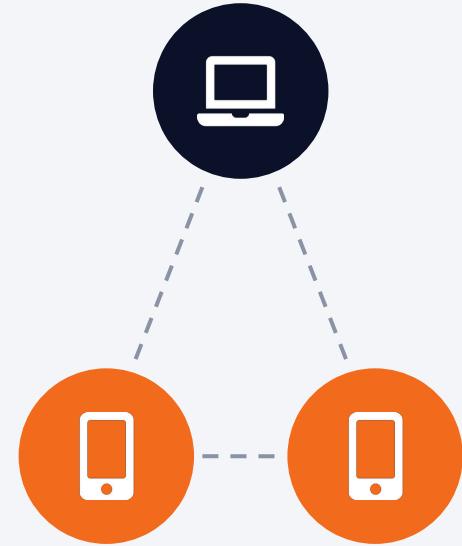


MeshAI

Run AI models your laptop can't, using the phone you already own.

Team	Maynards
Track	Developer Tools
Members	Prajwal, Yuvaraj
Problem statement	Developers with budget laptops cannot run newer open AI models without buying new hardware or sending private code to the cloud.



One private machine

Problem Statement

- New open models like gpt-oss-20b (12.1 GB) do not fit on a budget laptop. An 8 GB laptop has about 4 GB free.
- The phone cannot hold them alone either. A 16 GB phone has about 10 GB usable.
- A GPU upgrade often costs more than the laptop itself.
- Many companies do not allow code to be pasted into cloud AI.

Who faces this

Students and developers with a good phone and a modest laptop.

Small teams working on code they cannot share with cloud services.

Result: they settle for weaker models, or pay for cloud AI and give up privacy.

Model size: ggml-org/gpt-oss-20b-GGUF. Usable phone RAM: PowerInfer-2 (arXiv 2406.06282) measured 11 GB free on a 16 GB phone.

Proposed Solution

MeshAI pools the memory of your phone and laptop into one private machine, so models too big for either device can run across both.



Bigger models

Load models that neither device can hold alone, like gpt-oss-20b on an 8 GB laptop plus one phone.



Private and offline

Code and data stay on your own devices. No cloud account, no internet needed.



No new hardware

Uses the phone you already have instead of a new GPU or laptop.

The phone does most of the work: it holds the larger share of the model and is where you pair, watch and control the mesh.

What Makes It Different

Existing work	What it does	What is missing
llama.cpp RPC	Splits a model across machines	Manual setup with command line flags and IP addresses. No phone app.
Exo	Automatic device clusters	Built mainly for Macs and desktops, not Android phones.
prima.cpp	Research system for home clusters	Research code. Phones run it through Termux.
AI Doomsday Toolbox	Android phone clusters	Android only and manual. No laptop and phone mix.

Our USP

- Phone app that joins with a QR scan. No terminal, no IP addresses.
- A scheduler that decides when to split, and when not to use a device at all.
- One local API, so any coding tool works without changes.

Key Features



QR pairing

Scan a code on the laptop to join. Uses the phone camera.



Device profiling

Free RAM, CPU, GPU and NPU speed, heat, battery, link speed.



Smart scheduler

Runs on one device, splits the model, or runs jobs in parallel.



One local API

OpenAI compatible, so coding tools and scripts just connect.



Background jobs

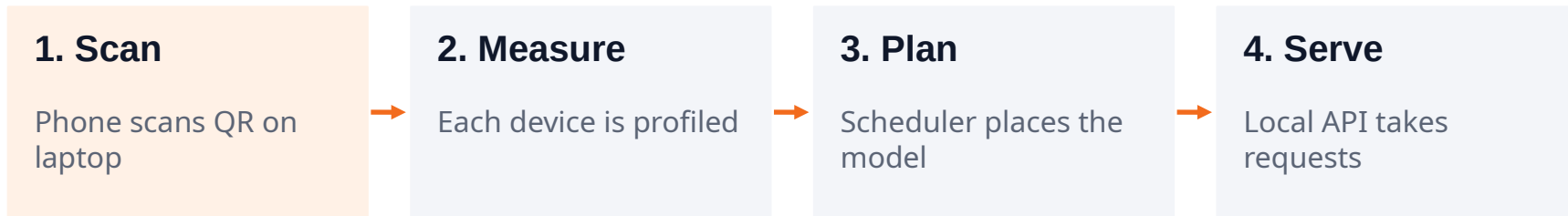
Queue long tasks like writing tests and come back to results.



Live phone dashboard

Layers held, tokens per second, temperature and battery.

Process Flow and Use Cases



Use cases: jobs where waiting a few minutes is fine and privacy matters



Write tests for a module



Review a branch before merging



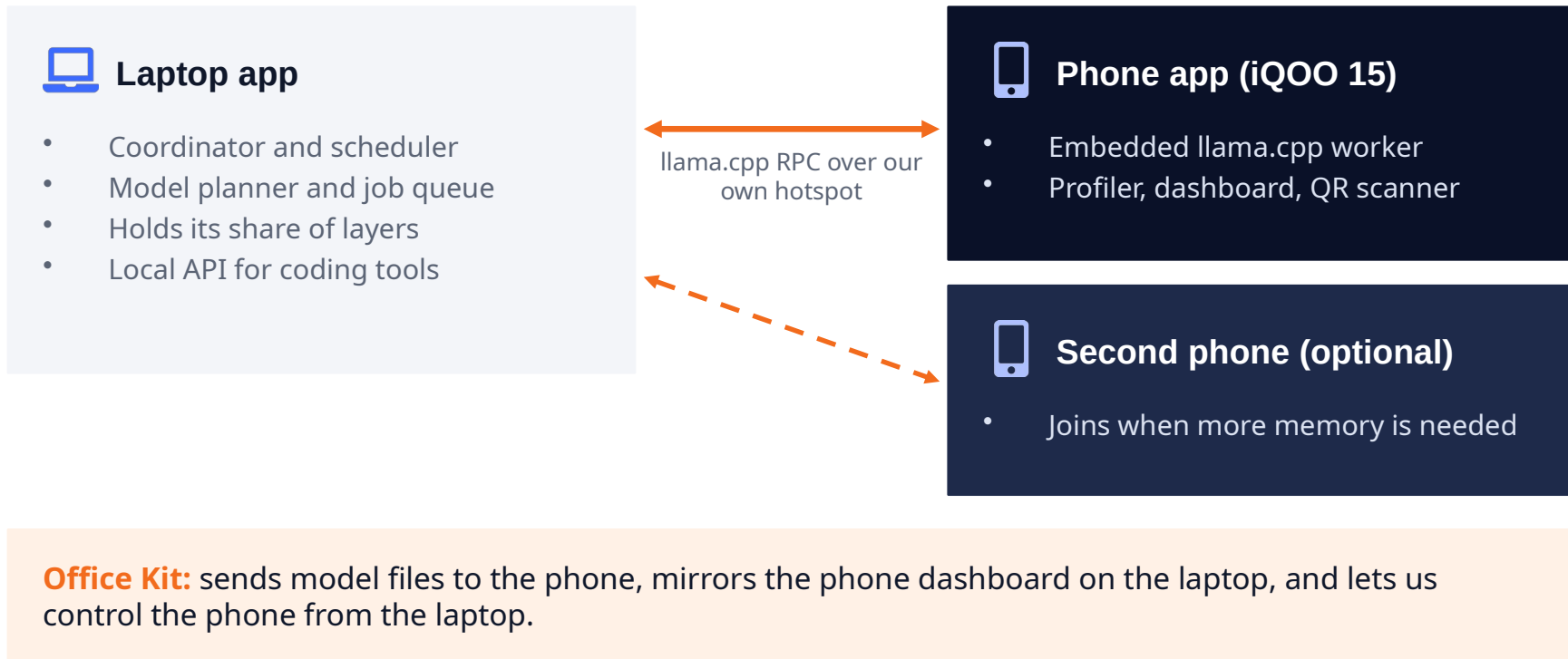
Document old code



Read large logs and crash reports

Start the job, keep working, and come back to the result. Fully offline.

Architecture

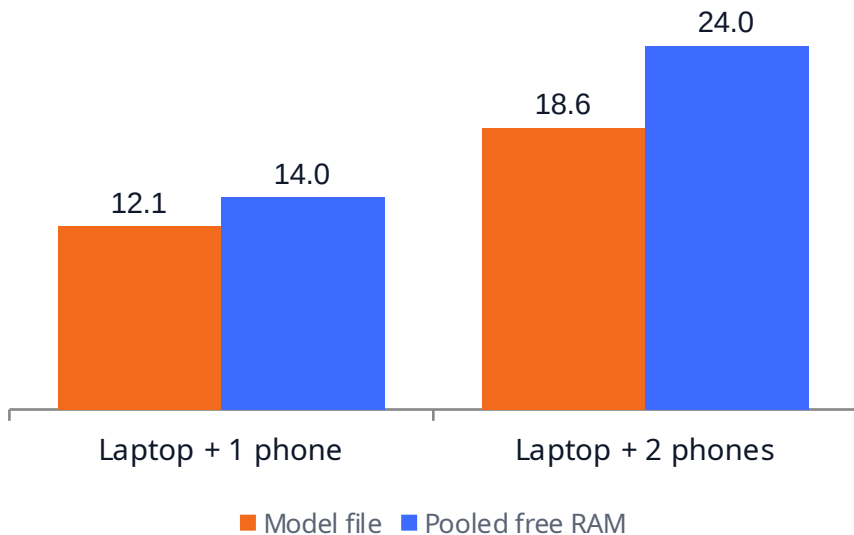


Tech Stack

Layer	Option A	Option B
Phone app	Kotlin with Jetpack Compose	Flutter with a native plugin
On device inference	llama.cpp built with the Android NDK (C++)	llama.cpp in Termux, for quick tests only
Phone compute	CPU (ARM)	Adreno GPU via OpenCL, Hexagon NPU (experimental)
Model splitting	llama.cpp RPC backend	None needed when the model fits one device
Laptop coordinator	Python (FastAPI)	Go or Node.js
Local API	llama.cpp server, OpenAI compatible	Thin proxy in the coordinator
Models (GGUF)	gpt-oss-20b (MXFP4)	Qwen3-Coder-30B-A3B (Q4_K_M), small models for tests
Pairing and link	QR code, our own hotspot	USB tethering, Wi-Fi Direct later
Phone and laptop bridge	vivo Office Kit	None

Technical Feasibility

Memory (GB): gpt-oss-20b, then Qwen3-Coder-30B-A3B



Why it works

- Phone memory is fast: about 85 GB/s on the iQOO 15 against 25 to 51 GB/s on a budget laptop.
- Newer MoE models use only 3.3 to 3.6B parameters per token, so each token reads about 2 GB.
- Only a few KB cross the network per token between devices.

Target: 3 or more tokens per second for gpt-oss-20b on a laptop and one phone. Measured on day one.

Pools assume an 8 GB laptop and 16 GB phones. Qualcomm 8 Elite Gen 5 brief, JEDEC DDR4, model cards, llama.cpp RPC discussion #9136.

Risks and Limits

Risk	What we do
Splitting is slower than one device	Split only when the model does not fit. Research agrees (prima.cpp, arXiv 2504.08791).
Network overhead	RPC lost 28 to 55% speed even on wired LAN (llama.cpp #22850). We target background jobs.
NPU support is experimental	CPU first, GPU and NPU only if they measure faster and correct (llama.cpp #25876).
Phone heats up	Heat aware scheduling, charging, a cap on the phone's share.
Venue Wi-Fi blocks devices	Our own hotspot, USB tethering as backup.
Android kills the worker	Foreground service with memory headroom.

We do not claim cloud speed, or that splitting beats a device that already fits the model.

Implementation Plan

Build At the hackathon

- Laptop and one iQOO 15 running gpt-oss-20b split
- Phone app: worker, QR pairing, profiler, dashboard
- Scheduler: one device or split
- Local API used by a coding tool
- One background job type end to end

Stretch If time allows

- Second phone and Qwen3-Coder-30B-A3B
- Parallel jobs across devices
- Adreno GPU backend if faster

Later After the hackathon

- Hexagon NPU backend
- Expert placement for MoE models
- Wi-Fi Direct, encrypted links

Impact and Benefits

33M

phones above ₹30,000 sold in India in 2025

7.3M

consumer PCs sold in India in 2025

9%

of Indian households own a computer

- Access: strong AI for people who own a good phone but not a powerful PC.
- Privacy: in 2023 Samsung banned staff from ChatGPT after source code was pasted into it. MeshAI keeps code on your devices.
- Cost: no GPU purchase and no cloud bills for repeated jobs.
- Scale: add another phone for a bigger model. Works with any GGUF model.

IDC and Counterpoint 2025 India (22% of 152M phones). NSO CAMS 2023 via Data For India. Bloomberg, 2 May 2023.

Research and References

- [prima.cpp](#): home device clusters over Wi-Fi. arXiv 2504.08791
- TPI-LLM: 70B on low resource devices. arXiv 2410.00531
- EdgeShard: split LLM inference at the edge. arXiv 2405.14371
- LinguaLinked: LLMs across Android phones. arXiv 2312.00388
- MELting Point: LLMs on phones. MobiCom 2024
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- [llama.cpp](#) RPC, OpenCL and Hexagon docs, github.com/ggml-org/llama.cpp
- [llama.cpp](#) issues #22850, #25876, discussion #9136
- [gpt-oss-20b](#) and [Qwen3-Coder-30B-A3B](#) model cards, huggingface.co
- Qualcomm Snapdragon 8 Elite Gen 5 product brief
- [iQOO 15 specifications](#), iqoo.com
- IDC, Counterpoint 2025. NSO CAMS 2023